because titles contain a significantly smaller number of words than abstracts, allowing us to obtain a text layer similar to the document layer in terms of number of edges by simply cutting out words present in less than 5 documents. It is indeed well known that the hSBM performs optimally when both layers have roughly the same size, otherwise the smaller layer is effectively ignored by the algorithm [21]. The filtered dataset hence consists of 181 605 documents and 15 381 different words, summing up to 590 215 document-to-document citation links and 1 396 830 document-to-word links.

To ensure the robustness of our results, we performed 100 iterations of the algorithm. Notice that the number of clusters and levels of granularity obtained is not fixed, but is automatically suggested by the algorithm. By running the algorithm multiple times, we aimed to capture the inherent variability and uncertainty in the Monte Carlo partitioning process. Subsequently, the consensus partition is calculated by maximizing the overlap with all the partitions from the 100 runs. Such consensus partition serves as a robust representation of the underlying structure within the analyzed data. More statistics comparing the single iterations of the hSBM and the consensus partition are shown in Fig. S2 in the SM.

Afterwards, keywords are assigned to each cluster to roughly represent the content and themes of the articles within them (for more details see Fig. S3, Fig. S4 and Table S4 in the SM, with related section). Keywords are chosen by looking at the most frequent words in the cluster, the most significant topics in the cluster according to the normalized mixture proportion [21], as well as the first 5 papers in the cluster according to different measures (see SM Section 2.1 for more details).

Step 3: Knowledge flows

In the third step of the pipeline, we want to better understand how the different groups of documents identified by the hSBM have influenced each other throughout history. To do so, we evaluate the knowledge flows between these groups, using article citations as proxy [23]. In particular, we compute the knowledge flow from one cluster a in one year Y_a to another cluster b in a future year Y_b. The computation takes into account the fraction of citations towards papers in a of the year Y_a from papers in b published in the year Y_b with respect to the fraction of citations towards a in Y_a from all papers published in Y_b, as well as the overall fraction of papers of a in Y_a. The citation network suffers indeed from a series of inherent biases: field size, typical number of citations in a field or a journal, typical number of references, age of the fields etc. This method de facto considers the number of citations with respect to a null model, resulting in a link weight which is effectively a z-score.

Mathematically, if a collection of papers is divided in a partition $\mathcal{P}$ of clusters such that different clusters do not overlap and altogether form the collection of papers, then we can define the knowledge flow units $C_{a\to b}(Y_a, Y_b)$ from papers in cluster $a \in \mathcal{P}$ published in the year Y_a to papers in cluster $b \in \mathcal{P}$ in a future year Y_b by counting how many citations have occurred from b to a in the two years, that is,

$$C_{a\to b}(Y_a, Y_b) = |\{(x,\, y) : x \in a,\, Y_x = Y_a,\, y \in b,\, Y_y = Y_b \text{ s.t. } y \text{ cites } x\}|. \tag{1}$$

As said before, we need to normalize this number with respect to a null model, so as to keep into account different sizes of clusters and different norms in citation practices. Hence, the knowledge flow

$K_{a\to b}(Y_a, Y_b)$ from a in year Y_a to b in year Y_b can be computed in the following way:

$$K_{a\to b}(Y_a, Y_b) = \begin{cases} 1 & \text{if} \quad \dfrac{C_{a\to b}(Y_a, Y_b)}{\sum_{c\in\mathcal{P}} C_{c\to b}(Y_a, Y_b)} \Big/ \dfrac{|x \in a : Y_x = Y_a|}{\sum_{c\in\mathcal{P}} |x \in c : Y_x = Y_a|} \geq 1 \\ 0 & \text{otherwise} \end{cases} , \tag{2}$$

After the normalization against the null model, knowledge flows can be indeed treated as z-scores. Hence, in Eq. (2) we consider a knowledge flow as significant (i.e., a binary value of 1) if higher than the threshold 1, and as not significant (i.e., 0) otherwise.

Therefore, we obtain a binary value for each pair of clusters and each pair of years. In other words, the collection of (knowledge flow) links between all pairs of clusters and years generates a temporal network of clusters, which we aggregate in different ways to facilitate the following analysis. In particular, we consider the average knowledge flow $K_{a\to b}(Y_a)$ from a cluster a to another b from a specific year Y_a as the average of the knowledge flows $K_{a\to b}(Y_a, Y_b)$ from cluster a to b from year Y_a to all years $Y_b > Y_a$, taking into account only years Y_b where there is at least one publication in b. Formally, this reads:

$$K_{a\to b}(Y_a) = K_{a\to b}(Y_a, \bullet) = \langle K_{a\to b}(Y_a, Y_b)\rangle_{\{Y_b > Y_a \,:\, \exists x \in b \text{ s.t. } Y_x = Y_b\}} \ . \tag{3}$$

This value represents, on a scale from 0 to 1, how much publications in cluster a in year Y_a have influenced the future of cluster b. Analogously, we define the average knowledge flow $K_{a\to b}(T)$ from cluster a to cluster b from a period of time T to the future by averaging $K_{a\to b}(Y_a)$ over all years Y_a in T in which there is at least one publication in a, that is,

$$K_{a\to b}(T) = \langle K_{a\to b}(Y_a)\rangle_{\{Y_a \in T \,:\, \exists x \in a \text{ s.t. } Y_x = Y_a\}} \ . \tag{4}$$

We can also measure the average influence in terms of knowledge flows from a cluster to all other clusters and vice-versa, as well as the average knowledge flow among all clusters, respectively as follows:

$$\begin{aligned} K_{a\to\bullet}(Y) &= \langle K_{a\to b}(Y)\rangle_b \ , \\ K_{\bullet\to a}(Y) &= \langle K_{b\to a}(Y)\rangle_b \ , \\ K_{\bullet\to\bullet}(Y) &= \langle K_{a\to b}(Y)\rangle_{a,b} \ . \end{aligned} \tag{5}$$

Here, $K_{a\to\bullet}(Y)$ refers to the average influence from papers in cluster a published in year Y towards all clusters in the future. On the opposite, $K_{\bullet\to a}(Y)$ refers to the average influence of the papers in all clusters in the year Y towards the future of cluster a. Finally, $K_{\bullet\to\bullet}(Y)$ refers to the average influence (towards the future) of all papers published in year Y.

3 Results

3.1 The decentralized evolution of (de)centralization

We start by analysing the number of papers mentioning *(de)centralization* over the years (see *The pipeline* section for more details), comparing it to the total number of academic outputs (papers, books etc.) produced in time, which is known to be exponentially increasing [24]. As shown in Fig. 1(a), the fraction of papers mentioning *(de)centralization* has been exponentially increasing in time since the

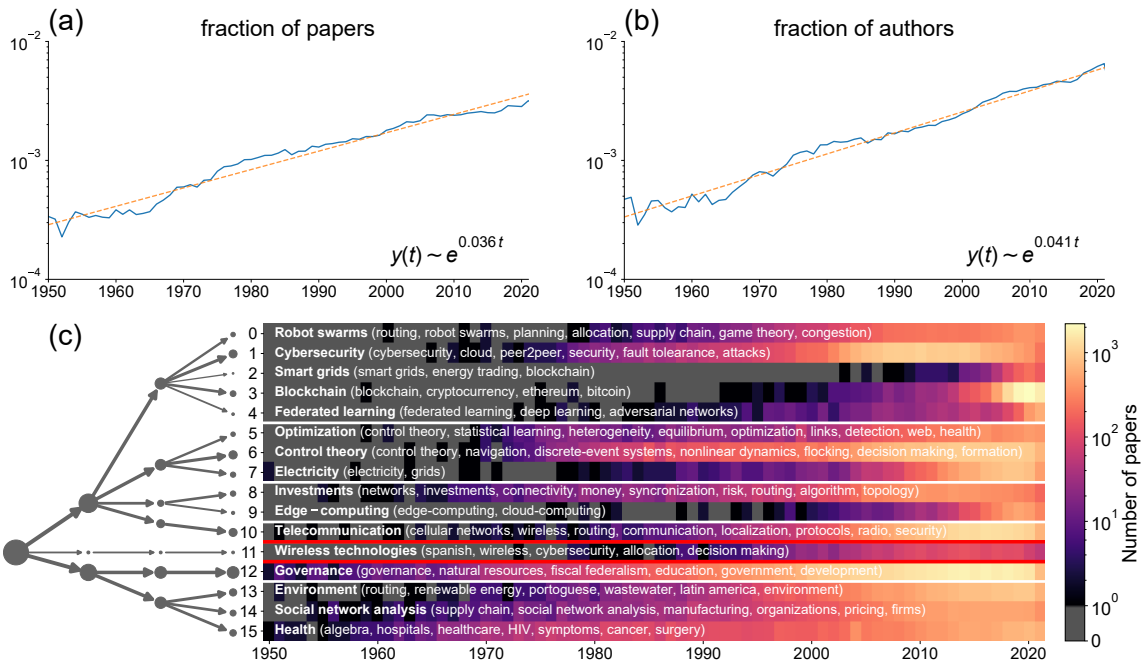

Figure 1: **The rising interest of academic literature towards *(de)centralization*. (a)** Fraction of papers mentioning *(de)centralization* in the Semantic Scholar corpus. **(b)** Fraction of authors mentioning *(de)centralization* in the Semantic Scholar corpus. Both fractions have been steadily increasing since the 1950s, showing growing interest in the topic. **(c)** Number of papers in the clusters at the 3rd level of the hierarchy in each year. Clusters are ordered respecting the hierarchical network on the left, in which node and link sizes are proportional to the total number of papers in the related cluster. In the heatmap, white lines individuate clusters belonging to the same cluster at the 2nd level of the hierarchy, while red lines divide different clusters at the 1st level. The representative keyword of the clusters at level 3 is reported in bold in the respective rows, while all the specific keywords identified at the 4th hierarchical level are shown within brackets. Clusters with less than 500 papers in total are not shown in the figure.

1950s, rising to around one paper every 315 in 2021. The growing interest in this topic is also reflected by the increasing number of authors involved in such academic research. Indeed, as shown in Fig. 1(b), the fraction of authors producing such research has risen exponentially by more than one order of magnitude, with almost one academic every 154 writing a paper mentioning the topic in 2021. This growth is also seen in terms of raw number of publications and authors, as shown in Fig. S1 in the SM, where we compare these numbers for the S2AG corpus and the *(de)centralization* dataset and find a stronger exponential rise for the latter. Notice that for both papers and authors there are some periods with a higher or lower increase in the fraction, showing spikes of interest at particular times. For example, in Fig. 1(a,b) we can see that between the late 1970s and the 1980s the growth rate was faster than the overall exponential fit.

In order to understand what has characterized the origins and evolution of the topic, we set to identify topics and clusters of papers in the dataset by using the hSBM algorithm [20, 21] described in *The pipeline* section. In the following analysis, we focus our attention only to years after 1950. Before this date there are only around 100 papers in our *(de)centralization* dataset. The very first is a political science one from 1851 on local self-governments versus centralized governments [25]. Among the others in this period, apart from around 50 papers that relate to *(de)centralization* in governments, organizations and states, we have detected 30 papers that are actually false positives of the selection process. Considering also how, in general, digitalization issues may have contributed to the small number of papers before 1950,